Phylomic

BIOINFORMATICS ANALYSIS & REPORTING

RNA-seq Differential Expression Analysis

Pulmonary fibrosis: single-dose bleomycin vs saline, mouse lung (GSE301216)

Prepared for research demonstration

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Executive Summary

Study. Public dataset GSE301216: bulk RNA-seq of mouse lung following a single intratracheal dose of bleomycin, a standard model of pulmonary fibrosis. **Comparison:** single-dose bleomycin (n = 14) vs vehicle controls sequenced in the same run (n = 10), 21 days after intratracheal administration.

Headline result. Bleomycin produced a large, reproducible transcriptional response. **6,410 genes were robustly differentially expressed** (3,365 up, 3,045 down), meaning significant at FDR < 0.05 with the same direction in both the primary analysis and an independent sensitivity analysis. The primary model identified 8,428 genes at FDR < 0.05; the robust set is the conservative figure used throughout this report.

Three biological themes

1. Fibrotic remodeling. The strongest signal in the dataset. The Hallmark epithelial-mesenchymal transition program was enriched among upregulated genes (129 of 194 genes, FDR 1.7×10^{-43}), alongside extracellular matrix organization. Core fibrosis genes all increased: Col1a1 3.9×, Col3a1 3.9×, Fn1 5.7×, Loxl2 3.1×, Mmp2 5.3×, Timp1 8.7× and Spp1 12×. Senescence-associated Cdkn1a (3.3×) and Cdkn2a (2.7×) also rose.

2. Inflammatory and immune remodeling. Upregulated genes were enriched for leukocyte migration (FDR 7.6×10^{-21}), interferon-γ response, complement and TNF-α/NF-κB signaling. Il6 increased 3.5×.

3. Loss of the airway cilia program. Downregulated genes were dominated by motile cilia machinery: axoneme assembly (FDR 8.5×10^{-33}), with the ciliated-cell regulator Foxj1 at 0.6×. This is consistent with airway epithelial injury, or with ciliated cells becoming a smaller share of the tissue as other cells infiltrate. Some enrichment terms carry reproductive labels ("sperm motility"); these reflect ciliary genes shared with sperm flagella, not reproductive biology.

> **QC finding: tissue-associated expression signature detected**
>
> Seven of 24 samples showed a coherent skeletal-muscle-associated expression program. Five canonical markers (Acta1, Myh1, Atp2a1, Tnnc2, Tnnt3) were confirmed using a curated ortholog annotation combining multiple databases. The signal was present in 4 of 10 controls and 3 of 14 treated samples. The continuous contamination score was not significantly associated with treatment (p = 0.084), although the group imbalance was retained as an explicit QC consideration. Removing flagged samples did not fully eliminate it, since residual low-level signal remained in two controls. A normalized contamination score was therefore included in the primary model, and an independent sensitivity analysis was run on the 17 unflagged samples.
>
> **Result:** the main biological conclusions were highly stable between the two analyses (fold-change correlation r = 0.97). A common alternative, simply removing the flagged samples, would have produced a false result: muscle genes appeared strongly downregulated by bleomycin (Acta1 24-fold lower, padj < 0.001), because low-level signal remained in two controls. In the covariate-adjusted model, these genes showed no treatment effect.

Confidence classification

Class	Genes	Meaning
Robust	6,410	Significant in both analyses, same direction
Borderline	2,505	Significant in one analysis only, same direction, small effects; most (1,995) lost significance in the smaller sensitivity analysis
Contamination-sensitive	51	Result changes materially depending on how contamination is handled; excluded from interpretation
Not assessable in sensitivity analysis	3	Too few reads to test in the 17-sample analysis; not interpreted

Key takeaways

1. Bleomycin induced a strong fibrotic program, reproducing every fibrosis marker highlighted in the original publication.
2. Immune and inflammatory remodeling accompanies fibrosis, led by leukocyte recruitment and interferon-γ signaling.
3. The airway cilia program is suppressed, pointing to epithelial injury.
4. A tissue-associated contamination signal was detected and explicitly controlled; the main biological conclusions were robust to this correction.

2. Experimental Design

Study context. GSE301216 comes from a published comparison of two bleomycin lung fibrosis models: a single-dose model and a repetitive-dose model. Each model was run as a separate experimental cohort with its own saline controls. This report analyzes the single-dose model only.

Animals and treatment. Male C57BL/6JRj mice (12 weeks old) received a single intratracheal dose of bleomycin (2.0 mg/kg) or saline vehicle. Lungs were collected 21 days after administration. RNA from the inferior lobe was sequenced as poly(A)-selected bulk RNA-seq.

Comparison analyzed

Group	Treatment	n (sequenced libraries)	Harvest
Control	Saline, intratracheal	10	Day 21
Bleomycin	Single dose, intratracheal	14	Day 21

Why 10 controls rather than 20. GEO lists 20 control samples for this series. They belong to two cohorts: 10 to the single-dose experiment and 10 to the repetitive-dose experiment. The two sets were sequenced in separate runs, and they differ technically: the repetitive-cohort controls detect about 4,000 fewer genes on average (18,664 vs 22,807). All 14 bleomycin samples were sequenced alongside the single-dose controls. Pooling all 20 controls would therefore mix a sequencing-run difference into the treatment comparison. Using the 10 same-run controls matches the original study design, in which each model was compared against its own controls.

Samples not analyzed. The repetitive-model cohort (10 treated, 9 baseline, 10 controls) is outside the scope of this comparison and was excluded.

Sample identity. Each count-matrix column was matched to its GEO sample record through the library name recorded in the GEO metadata (53 of 53 samples matched one-to-one). The full sample table, with GEO accessions, is in the Appendix.

Replication. With 10 and 14 biological replicates, the comparison provides substantial replication for differential expression analysis and supports the predefined sensitivity analysis.

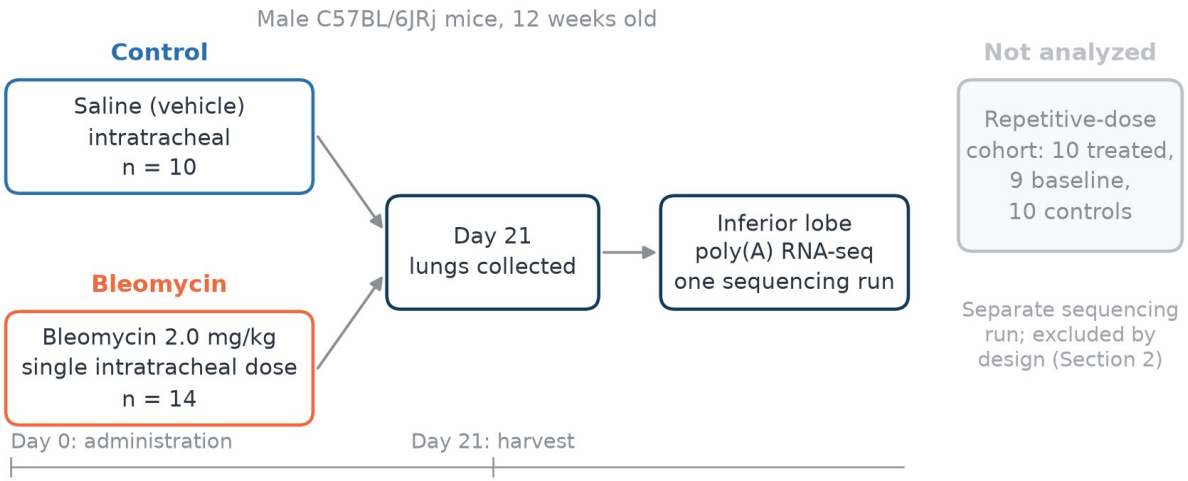


Figure 1. Study design. Mice received a single intratracheal dose of bleomycin (2.0 mg/kg, n = 14) or saline (n = 10), and lungs were collected 21 days later for RNA-seq of the inferior lobe. The repetitive-dose cohort, sequenced in a separate run, was not analyzed.

3. Data Quality

QC verdict. All 24 libraries passed technical QC and were retained. Two sample-level issues were found and handled explicitly: (1) controls from a separate sequencing run, excluded by design (Section 2); and (2) a tissue-associated expression signature in 7 samples, controlled statistically. Neither affects the main conclusions.

3.1 Sequencing depth and gene detection

Libraries were evenly sequenced across groups, with no low-depth samples.

	Controls (n = 10)	Bleomycin (n = 14)
Reads per library, mean (range)	9.6M (7.5 to 10.4M)	9.7M (7.4 to 11.2M)
Genes detected, mean (range)	22,807 (22,179 to 23,322)	22,925 (21,308 to 23,831)

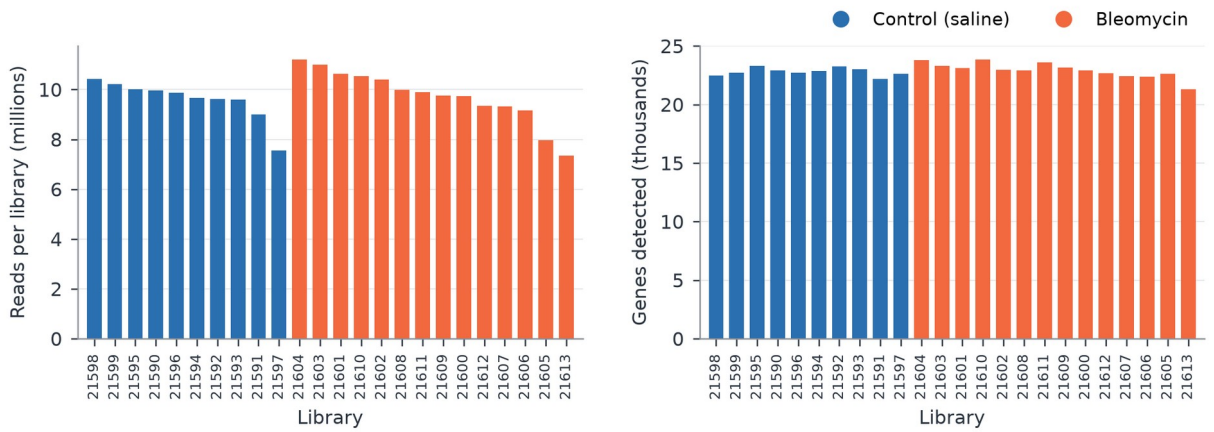


Figure 2. Library size and genes detected per sample, colored by group.

3.2 Sample integrity

No sample showed signs of being an outlier or a technical duplicate. Within-group correlations were uniformly high (median r = 0.982), and no pair of libraries was more similar than expected for independent biological replicates.

3.3 Overall sample structure

Treatment was clearly visible in the data: control and bleomycin samples separated completely along the second principal component (17% of variance), with no overlap. However, the largest source of variation (PC1, 58%) did not correspond to treatment. It split 7 samples, from both groups, away from the rest. Section 3.4 identifies its source.

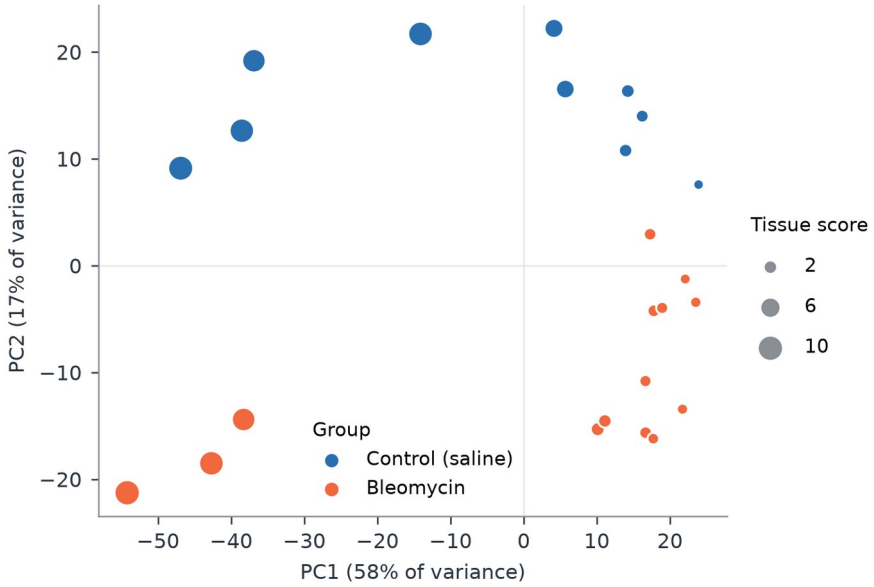


Figure 3. Principal component analysis of the 24 samples, colored by group, with point size scaled to the tissue-contamination score. PC2 separates the treatment groups; PC1 separates the 7 samples carrying the tissue signature.

3.4 QC finding: a skeletal-muscle-associated signature

Detection. The genes driving PC1 included canonical skeletal-muscle markers. Five were confirmed by curated annotation (Acta1, Myh1, Atp2a1, Tnnc2, Tnnt3), and a broader predefined 13-gene muscle panel behaved as a single coherent program (mean inter-marker correlation r = 0.92). Expression was effectively all-or-none: high in 7 samples, low-level in 2, and near zero in the remaining 15. That pattern is more consistent with differences in tissue composition than with a graded biological response.

Accompanying signal. In 6 of the 7 samples, the muscle signal was accompanied by squamous epithelial keratins (Krt4, Krt13). Together these are consistent with a small amount of adjacent non-lung tissue included at dissection, but that cannot be established from expression data alone. Markers of lung identity (surfactant genes) were unaffected in every sample.

Balance between groups. The signature was present in 4 of 10 controls and 3 of 14 treated samples. The continuous contamination score was not significantly associated with treatment (p = 0.084), although the group imbalance was retained as an explicit QC consideration.

Why it matters. Because the signal was more common in controls, it had the potential to bias the treatment comparison. In an unadjusted analysis, muscle genes showed no treatment effect; the high variability of the signal kept these estimates close to zero and far from significance.

Handling. Samples were not excluded on this basis. Instead:

1. A contamination score (mean normalized expression of the 13-gene panel) was included as a covariate in the primary model.
2. An independent sensitivity analysis was run after removing the 7 flagged samples (6 controls vs 11 treated). Flagged samples were identified by a natural break in the score distribution.

Evidence the correction worked. Sample exclusion alone was not sufficient. Two remaining controls carried low-level muscle signal, and in the sensitivity analysis Acta1 still appeared 24-fold lower in bleomycin lungs (padj < 0.001). In the covariate-adjusted primary model, the same gene showed no treatment effect (0.93×, not significant). Genes whose result depended on contamination handling (51 in total) were classified as contamination-sensitive and excluded from interpretation. Beyond these, conclusions were highly stable across the two analyses (fold-change correlation r = 0.97).

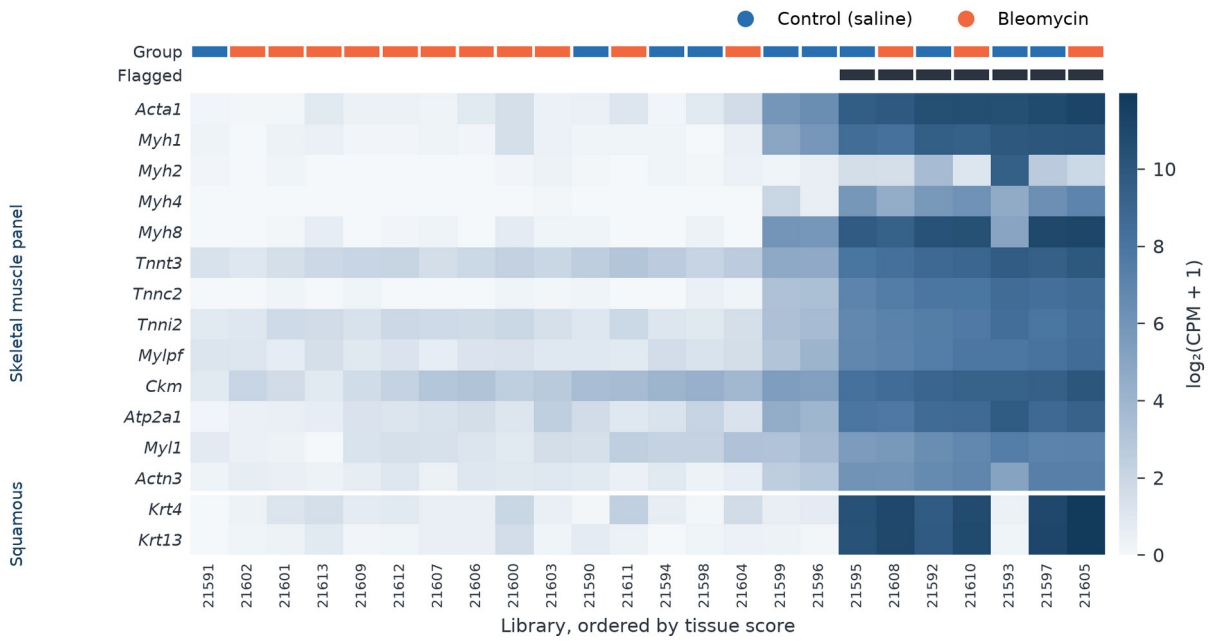


Figure 4. Expression of the skeletal-muscle and squamous marker panels across all 24 samples, ordered by tissue-contamination score. Flagged samples are marked above the heatmap.

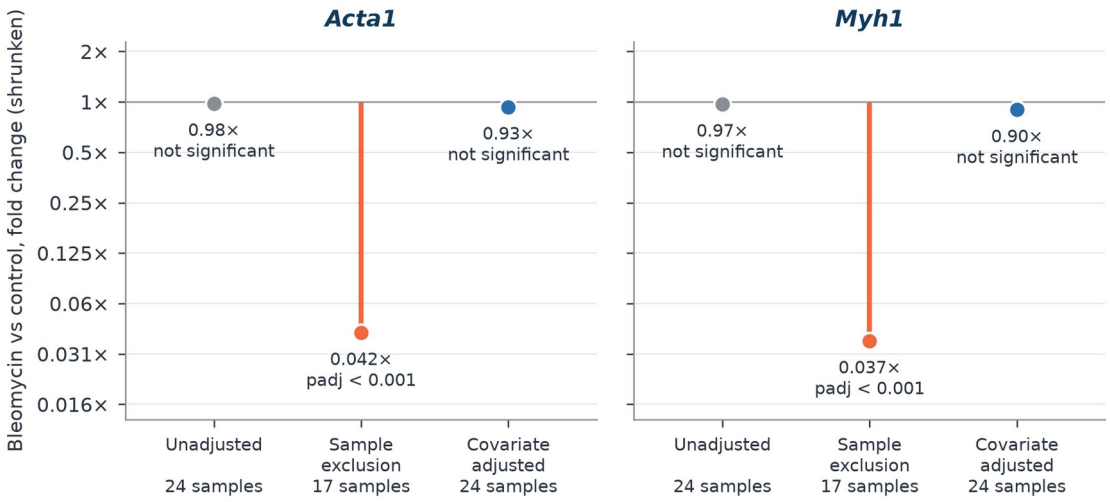


Figure 5. Removing flagged samples, the most common correction, created a false treatment effect; modeling the contamination did not.

3.5 Additional tissue-associated signals

Cardiac muscle genes (Myh6, Tnnt2, Myl7) varied between samples and were lower on average in bleomycin lungs (about 0.5×). Mouse pulmonary veins contain cardiac muscle, so variation in how much of this tissue each sample contains is a plausible explanation. None of these genes reached the robust threshold, and they are not interpreted as treatment effects.

Keratins associated with basal or squamous epithelium (Krt4, Krt5) were increased in bleomycin lungs. This pattern is consistent with injury-associated epithelial remodeling, but because these genes overlap the tissue-associated signature described in Section 3.4, they are not interpreted as independent treatment effects in this report.

4. Differential Expression

4.1 How differential expression was tested

Gene expression was compared between bleomycin and control lungs using DESeq2, a standard statistical method for RNA-seq count data. Genes with fewer than 10 total reads across all samples were removed before testing (24,818 genes tested). A gene was called differentially expressed at a false discovery rate (FDR) below 0.05, corrected for testing thousands of genes at once. No minimum fold change was imposed, so the gene lists capture small but consistent changes as well as large ones.

Two analyses were run, as described in Section 3.4:

- Primary analysis: all 24 samples, with the tissue-contamination score included in the model.
- Sensitivity analysis: the 17 samples without the tissue signature, with no contamination term.

4.2 Result classifications

Every gene significant in either analysis was assigned one of four result classifications. These classifications describe how consistently a result held up across the two analyses. They are not significance categories: every gene in all four classes passed the same FDR threshold in at least one analysis.

Classification	Genes	Definition
Robust	6,410	Significant in both analyses, with the same direction of change
Borderline	2,505	Significant in one analysis only, same direction, small effect (median 1.2-fold). Most (1,995) were significant only in the primary analysis, as expected when the sensitivity analysis has fewer samples
Contamination-sensitive	51	The estimated effect changed substantially (in direction, or by more than 2-fold) depending on how the tissue signature was handled
Not assessable in sensitivity analysis	3	Significant in the primary analysis but too few reads to be tested in the 17-sample analysis

Interpretation in this report is restricted to Robust genes. Borderline genes are provided in full in the results tables, since many likely reflect real but modest effects. Contamination-sensitive and not assessable genes are listed for transparency and not interpreted.

4.3 Scale and size of the response

The primary analysis identified 8,428 genes at FDR < 0.05. Of these, 6,410 were Robust (3,365 increased, 3,045 decreased). The two analyses agreed closely on the size of each change (fold-change correlation r = 0.97).

Most changes were moderate (median 1.4-fold among Robust genes), but a substantial group changed strongly:

Robust genes	Increased	Decreased
≥ 2-fold	833	326
≥ 4-fold	235	18

Large increases were far more common than large decreases. This fits a tissue in which new or expanding cell programs (such as activated fibroblasts and infiltrating immune cells) appear against a background of more modest losses. Section 5 examines what those programs are.

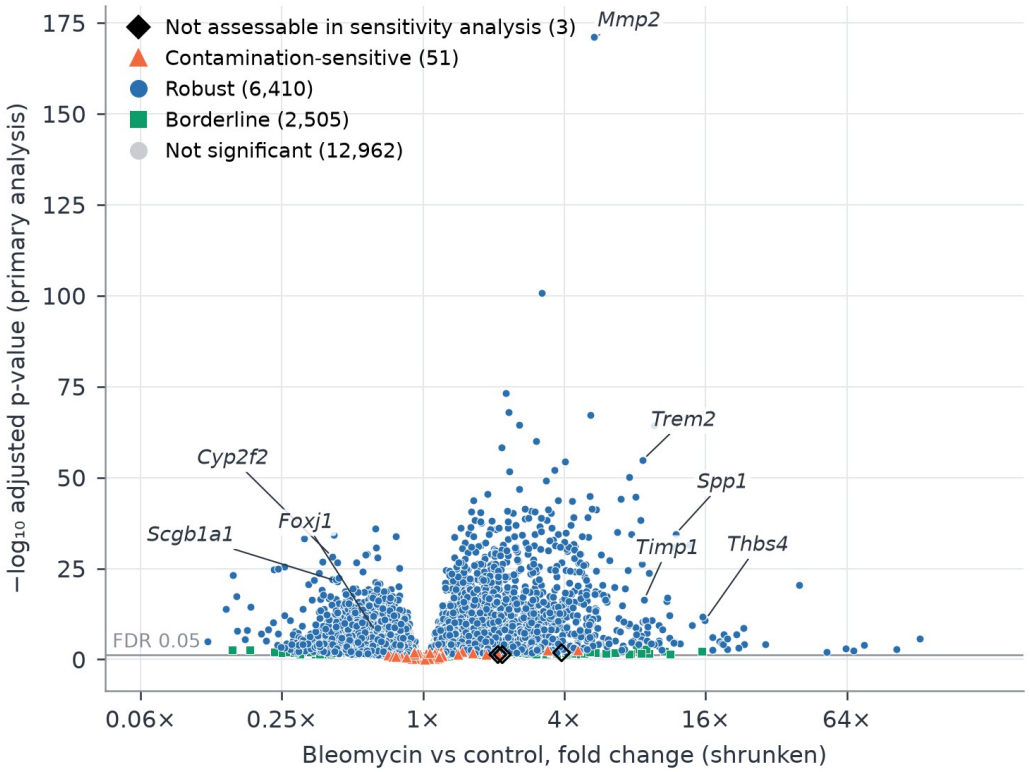


Figure 6. Volcano plot of the primary analysis, with points colored by result classification. Fold changes are shrunken estimates.

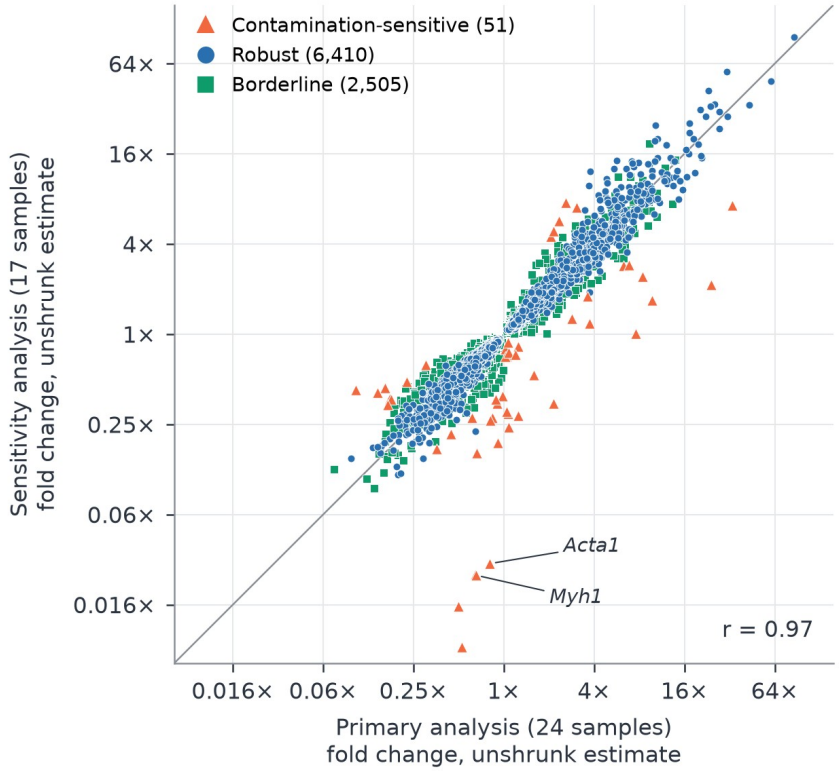


Figure 7. Fold changes in the primary and sensitivity analyses for all genes significant in either analysis (r = 0.97), colored by classification. Unshrunk estimates are shown, because the classification and the correlation are defined on them.

4.4 Most strongly supported changes

Top Robust genes changing at least 2-fold (mean normalized count ≥ 100, ranked by FDR). Genes associated with the tissue signatures in Section 3 are excluded (exclusion list: Myh6, Tnnt2, Myl7, Nppa, Tnnc1, Krt4, Krt5, Krt13, Krt14, Krt6a, Sprr3, Tgm3, Crnn, Pkp1; also Gm*, Ig*, *Rik).

Increased genes

Gene	Fold change	FDR
Mmp2	5.3×	9.3×10^{-172}
Ndnf	3.2×	1.7×10^{-101}
Ccng1	2.3×	7.3×10^{-74}
Sparc	2.3×	1.4×10^{-68}
Ctss	5.2×	6.9×10^{-68}
Slamf7	2.6×	3.8×10^{-65}
Psrc1	9.6×	6.3×10^{-65}
Mpeg1	3.0×	1.1×10^{-60}
Bgn	2.2×	4.9×10^{-59}
Trem2	8.6×	1.8×10^{-55}

Decreased genes

Gene	Fold change	FDR
Slc16a11	0.31×	6.3×10^{-34}
Cyp2f2	0.41×	7.5×10^{-29}
Pon1	0.37×	1.6×10^{-27}
Lypd2	0.26×	3.1×10^{-26}
Ttll6	0.41×	8.6×10^{-23}
Ttc16	0.42×	9.7×10^{-23}
Nrn1	0.34×	1.5×10^{-22}
Scgb1a1	0.43×	3.3×10^{-22}
Chad	0.32×	2.5×10^{-21}
Tymp	0.37×	7.0×10^{-20}

Complete results for all tested genes, with classifications, are provided in the Appendix and the accompanying results files.

4.5 Agreement with the original publication

The original study reported 5,577 differentially expressed genes for this comparison (2,718 increased, 2,859 decreased), also using DESeq2 at FDR < 0.05. This analysis identified more genes in the primary model (8,428) and a similar number in the Robust set (6,410), with the same near-even balance of increases and decreases. Differences in gene filtering, control selection and annotation version can each shift these totals, and the published gene-level results were not available for direct comparison. Every fibrosis, inflammation and senescence marker highlighted in the publication was reproduced (Section 5).

5. Biological Interpretation

How to read this section. All interpretation is based on Robust genes (Section 4.2). Fold changes are shrunken estimates. This is bulk tissue RNA-seq: each measurement is the average across every cell in the sample. A change in a gene's expression can reflect cells changing their activity, or a shift in which cells are present, and bulk data alone cannot always separate the two. Where the pattern points more toward one explanation, that is stated as an interpretation, not a finding.

5.1 Fibrotic remodeling

Fibrosis is the dominant signal in the dataset. The Hallmark epithelial-mesenchymal transition program, which in practice captures matrix-producing mesenchymal cells, was the most strongly enriched gene set among increased genes (129 of 194 genes, FDR 1.7×10^{-43}). Supporting GO terms included collagen fibril organization (FDR 4.0×10^{-11}) and wound healing (FDR 4.9×10^{-11}).

Matrix production and remodeling. Collagens (Col1a1 3.9×, Col3a1 3.9×, Col5a1 2.8×), fibronectin (Fn1 5.7×) and collagen cross-linking enzymes (Loxl2 3.1×, Lox 2.7×) all increased, alongside matrix-remodeling proteases and their inhibitor (Mmp2 5.3×, Timp1 8.7×).

Activated fibroblast programs. The largest increases came from genes associated with activated or injury-responsive fibroblasts: Thbs4 16×, Cemip 16×, Tnc 12× and Grem1 11×. Cthrc1, a gene associated with pathological fibroblast programs in lung fibrosis, rose 2.8×. At the same time, genes typical of healthy alveolar fibroblasts declined (Inmt 0.35×, Ces1d 0.63×). Together, this pattern is consistent with a shift in the fibroblast population from a homeostatic toward an activated state.

Two notes for interpretation.

- The classic myofibroblast marker Acta2 (α-smooth muscle actin) was unchanged. By day 21, matrix-producing fibroblasts may not be predominantly α-SMA-high, and Acta2 is also highly expressed by vascular and airway smooth muscle, which would dilute any fibroblast-specific change in bulk tissue.
- TGF-β signaling, a central driver of fibrosis, showed modest transcriptional evidence (Tgfb1 1.3×; Hallmark TGF-β signaling FDR 0.004). TGF-β activity is regulated largely at the protein level, so a small change in transcript does not indicate a weak pathway.

5.2 Immune and macrophage remodeling

Increased genes were enriched for leukocyte migration (FDR 7.6×10^{-21}), interferon-γ response (FDR 4.5×10^{-14}), complement (FDR 1.5×10^{-12}) and TNF-α/NF-κB signaling (FDR 1.9×10^{-11}). Il6 increased 3.5×.

Macrophages are the clearest immune signal. Genes characteristic of a profibrotic macrophage state rose sharply: Spp1 12×, Mmp12 14×, Trem2 8.6× and Gpnmb 7.7×, with broader macrophage markers also increased (Cd68 4.0×, Ctss 5.2×, Itgax 3.6×). This combination resembles SPP1- and TREM2-high macrophage programs reported in fibrotic lungs. Markers of resident alveolar macrophages changed far less (Siglecf 1.4×, Chil3 2.1×, Ear2 2.0×; Marco unchanged), which suggests the increase comes mainly from macrophages in an activated or recruited state rather than from expansion of the resident population.

Lymphocytes: a label that needs care. GO terms such as "B cell activation" were enriched, but B-cell marker genes decreased slightly (Cd19 0.66×, Cd79a 0.72×), and T-cell markers changed little (Cd3e and Cd8a unchanged; Cd4 1.6×). The enrichment is driven by immune-activation genes shared across many cell types, not by an expansion of B cells. The one clear lymphoid-lineage increase was Jchain (3.0×), a marker of antibody-secreting plasma cells.

5.3 Airway and alveolar epithelium

Loss of the airway cilia program. Decreased genes were dominated by motile cilia machinery: axoneme assembly (65 of 81 genes, FDR 8.5×10^{-33}) and cilium movement (FDR 1.4×10^{-29}). Ciliated-cell genes declined consistently (Foxj1 0.63×, Dnah5 0.59×, Tppp3 0.50×). Some enriched terms carry reproductive labels, such as "sperm motility"; these reflect cilia genes shared with sperm flagella, not reproductive biology.

Club cells declined in parallel. The secretory club-cell program fell to a similar degree (Scgb1a1 0.43×, Cyp2f2 0.41×, Scgb3a2 0.46×, Cldn10 0.47×). The coordinated loss of both major airway epithelial programs is consistent with injury to the airway epithelium, reduced airway representation in fibrotic tissue, or both.

Alveolar epithelium. Markers of alveolar type 2 cells, the surfactant-producing cells, were unchanged (Sftpc, Sftpb, Lamp3), and most type 1 cell markers were stable. Genes associated with injury-induced transitional epithelial states increased (Cldn4 2.8×, Sfn 1.9×, Lgals3 1.9×). This is consistent with ongoing alveolar repair at day 21, although these genes are not exclusive to epithelium, and bulk data cannot confirm the cell states directly.

5.4 Cell stress, senescence and proliferation

Senescence-associated and p53-related genes increased, consistent with the original publication: Cdkn1a 3.3×, Cdkn2a 2.7×, Gdf15 4.7× and Serpine1 3.3×. The Hallmark p53 pathway was enriched (FDR 3×10^{-6}).

Proliferation programs also increased, including E2F targets (FDR 3.0×10^{-18}), G2/M checkpoint genes (FDR 1.1×10^{-16}), Mki67 3.9×, Top2a 4.0× and Cdk1 4.7×. Their coexistence in bulk tissue does not imply that the same cells are simultaneously proliferating and senescent. The pattern is more consistent with different cell populations contributing to the two programs.

5.5 What this analysis can and cannot establish

Can establish:

- A robust, reproducible fibrotic, immune and epithelial transcriptional response at day 21, stable across two analytical approaches.
- Every fibrosis, inflammation and senescence marker highlighted in the original publication.

Cannot establish:

- Which cells drive each change. Bulk RNA-seq averages across cells; the cell-type interpretations above are inferences from marker genes. Single-cell or spatial data, or histology, would be needed to confirm them.
- Protein-level or functional changes. Transcript changes do not always translate proportionally to protein or activity (for example TGF-β).
- Timing. A single time point (day 21) captures one stage of an evolving injury-repair process.

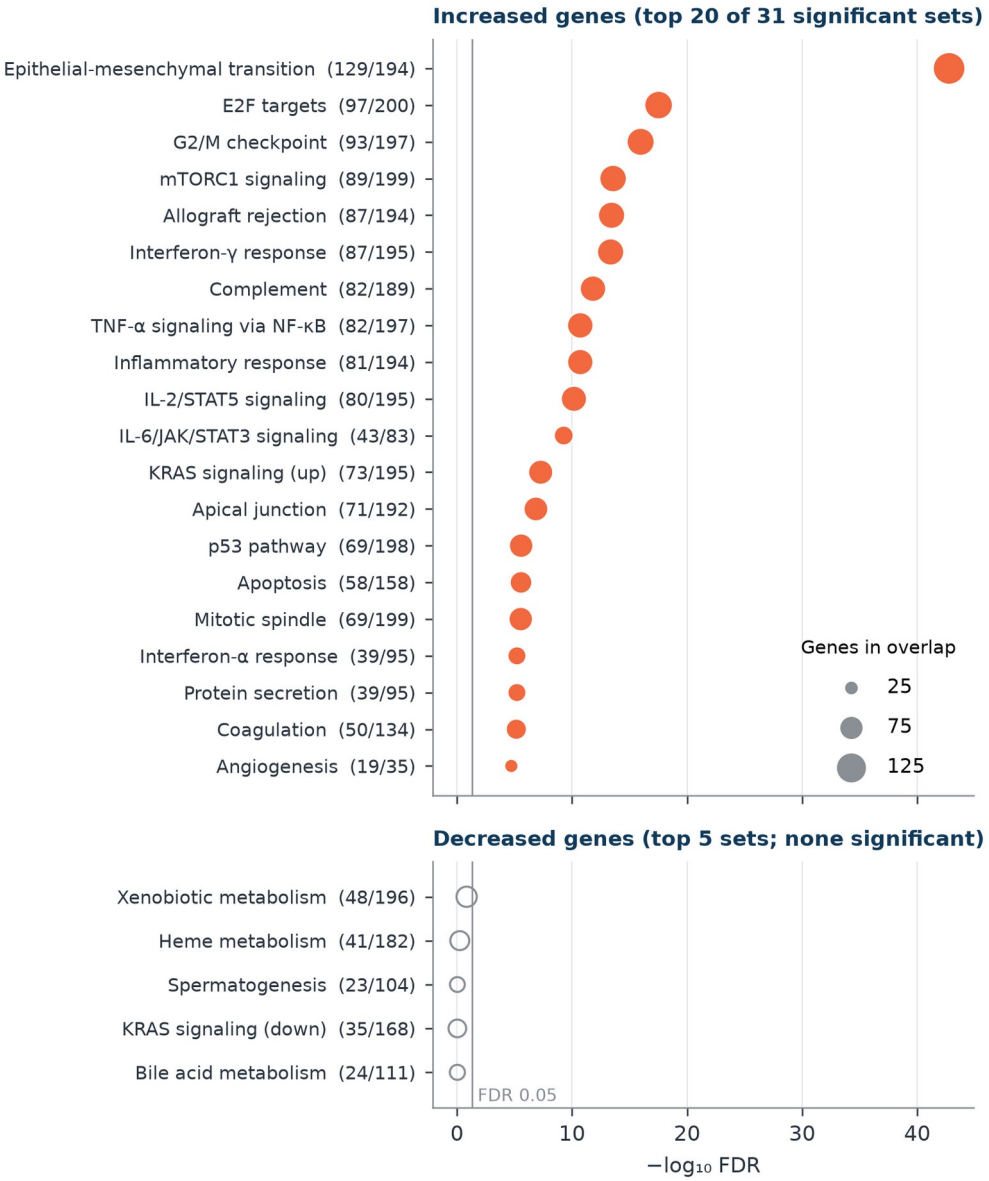


Figure 8. Hallmark gene sets enriched among increased genes (top 20 of 31 significant sets), and the top sets for decreased genes, none of which reached significance. Point size shows the number of overlapping genes; labels give the overlap and the set size.

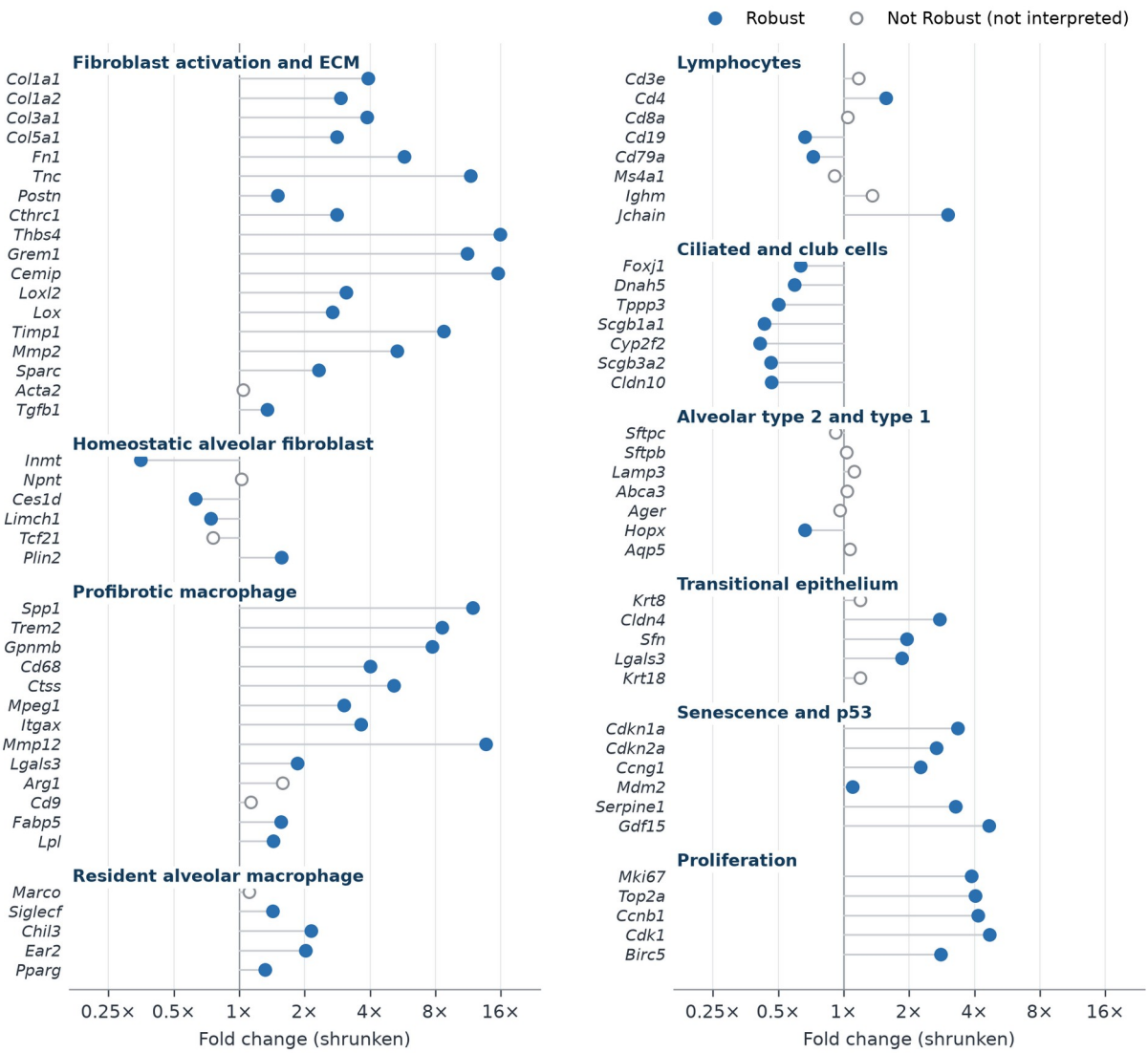


Figure 9. Fold changes (shrunken, primary analysis) for marker genes of the main lung cell programs. Filled points are Robust genes; open points did not meet the Robust criteria and are not interpreted.

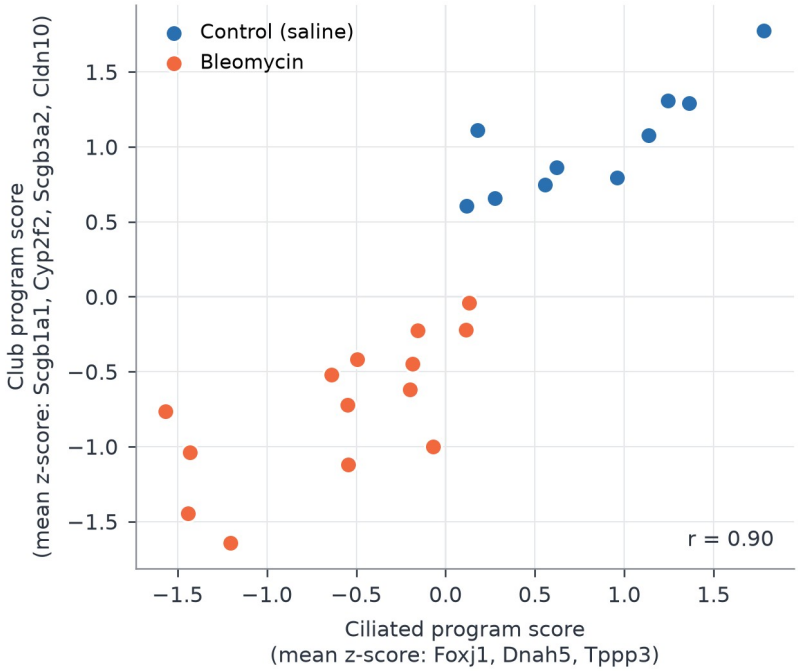


Figure 10. Coordinated decline of the ciliated and club-cell programs. Each point is one sample; each score is the mean z-score of the listed genes.

6. Key Figures

Curated set for reuse in manuscripts/presentations, with standalone captions: Figure 3 (PCA), Figure 5 (three-model comparison), Figure 6 (volcano), Figure 8 (Hallmark dot plot), Figure 9 (marker panels).

7. Methods

7.1 Data source

Raw gene-level read counts were obtained from the NCBI Gene Expression Omnibus, series GSE301216 (file GSE301216_BLEO-IPF_counts.txt.gz), with sample annotations from the series matrix. As reported by the data submitters:

- RNA was extracted from the inferior lung lobe (NucleoSpin RNA, Macherey-Nagel).
- Libraries were prepared with the NEBNext Ultra II Directional RNA Library Prep Kit (poly(A) selection) and sequenced on an Illumina NextSeq 500 (75 cycles).
- Reads were aligned and counted with STAR v2.7.0f against the GRCm38 mouse genome, Ensembl release 96.

No realignment was performed. Count-matrix columns were matched to GEO sample records through the library name recorded in each sample's GEO description (53 of 53 matched one-to-one).

7.2 Sample selection

The analysis included the 10 saline controls and 14 single-dose bleomycin samples sequenced in the same library run (Section 2). The 10 controls from the repetitive-dose cohort and all repetitive-model samples were excluded.

7.3 Quality control

- Library metrics: library size and number of detected genes (count > 0) per sample.
- Sample structure: PCA on $\log_2(\text{CPM} + 1)$ of the 500 most variable genes, among genes with ≥ 10 counts in ≥ 5 samples. Pearson correlations between samples on the same scale.
- Duplicate screen: pairwise correlations recomputed after excluding genes correlated with the contamination score ($|r| \geq 0.3$), and compared against the within-group distribution.

Tissue-signature score. A skeletal-muscle panel of 13 genes was defined: Acta1, Myh1, Myh2, Myh4, Myh8, Tnnt3, Tnnc2, Tnni2, Mylpf, Ckm, Atp2a1, Myl1, Actn3. The per-sample score was $\log_2$ of the mean CPM across the panel, plus 1. For modeling, the score was standardized (mean 0, SD 1). Samples were flagged at a score ≥ 6, which falls in a natural break in the distribution (observed values 4.8 and below vs 8.0 and above). The association between score and treatment was tested with a two-sided Mann-Whitney U test.

7.4 Differential expression

Differential expression was tested with PyDESeq2 (v0.5.4), a Python implementation of DESeq2. It used median-of-ratios normalization, a negative binomial GLM, the Wald test, Cook's distance outlier refitting, and independent filtering ($\alpha = 0.05$). P-values were adjusted by the Benjamini-Hochberg method. Genes with fewer than 10 total counts across the samples included in a given model were removed before fitting. No fold-change threshold was applied.

Model	Samples	Design	Genes tested	Role
Primary	24 (10 vs 14)	~ tissue score + condition	24,818	Main analysis
Sensitivity	17 (6 vs 11)	~ condition	23,390	Robustness check
Unadjusted	24 (10 vs 14)	~ condition	24,818	Comparison only (Figure 5)

7.5 Result classification

Genes significant (FDR < 0.05) in the primary or sensitivity model were classified using the unshrunk model estimates:

- Robust: significant in both models, with the same sign of $\log_2$ fold change.
- Contamination-sensitive: significant in at least one model, with opposite signs or an absolute difference in $\log_2$ fold change ≥ 1 between models.
- Borderline: significant in one model only, with the same sign and an absolute difference in $\log_2$ fold change < 1.
- Not assessable in sensitivity analysis: significant in the primary model but not tested in the sensitivity model because it fell below the read-count filter.

7.6 Effect-size estimates

For reporting, $\log_2$ fold changes for the condition coefficient were shrunk using PyDESeq2's lfc_shrink, which applies an adaptive Cauchy prior following the apeglm approach. Shrunken estimates are used for all fold changes reported in the text, tables and figures. Significance calls and result classifications are based on the unshrunk model results and are not affected by shrinkage.

7.7 Gene annotation and enrichment

- Annotation: mouse Ensembl gene IDs were mapped to gene symbols and to human orthologs using the babelgene ortholog database (v22.9). Where a mouse gene had multiple human orthologs, the one supported by the most source databases was used, with ties broken alphabetically by human gene symbol.
- Gene sets: Hallmark (H) and GO Biological Process (C5 GO:BP) collections from MSigDB v7.5.1, obtained through the msigdbr package (v7.5.1).
- Test: over-representation analysis by hypergeometric test, run separately for increased and decreased Robust genes. The background was all tested genes with a human ortholog (15,522). Gene sets with 15 to 500 genes after background restriction, and at least one overlapping gene, were tested. P-values were adjusted by the Benjamini-Hochberg method within each collection.

7.8 References

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- Benjamini Y, Hochberg Y. Controlling the false discovery rate: a practical and powerful approach to multiple testing. J R Stat Soc B 1995;57:289 to 300.
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- Bousamaki J, Rørbeck EA, Petersen AG, Korntner SH, Pors SE, Redente EF, Oró D, Salinas CG, Olsen RW, Simonsen U, Feigh M, Hansen HH. Distinct lung functional, histological and cell senescence signatures in the single and repetitive bleomycin mouse models of idiopathic pulmonary fibrosis. Physiological Reports 2025;13:e70560. doi:10.14814/phy2.70560.

8. Reproducibility

8.1 Inputs

File	Source	MD5 checksum
GSE301216_BLEO-IPF_counts.txt.gz	GEO GSE301216, supplementary file	34ef3b526f0b05a673f1bb0bb89c8bef
GSE301216_series_matrix.txt.gz	GEO GSE301216, series matrix	9e200d6e68b9cfd0481087c9b7154eb5

8.2 Reference resources (pinned)

Resource	Version	Source commit
babelgene ortholog database	22.9	ab61c529e31ce0bc1d0c17ef9c18bfa1bed41eb6
MSigDB via msigdbr	7.5.1	b396ab519fc522667383e0a80b75d9b53675b4d3

8.3 Software

Python 3.12.3; PyDESeq2 0.5.4; anndata 0.13.4; pandas 3.0.2; NumPy 2.4.4; SciPy 1.17.1; statsmodels 0.15.0; pyreadr 0.5.6.

The analysis contains no random sampling. The pinned inputs, reference resources, software versions and scripts are sufficient to reproduce the reported analyses and results, subject to the execution environment.

8.4 Outputs delivered

File	Contents
de_primary.csv	Primary model (24 samples, tissue score as covariate): all tested genes, unshrunk statistics and shrunken $\log_2$ fold change
de_sensitivity.csv	Sensitivity model (17 samples without the tissue signature), same columns
de_unadjusted.csv	Unadjusted model (24 samples, no covariate), for comparison only
de_classified.csv	Primary and sensitivity results side by side, with the result classification of every gene and its human ortholog
enrichment_{hallmark,gobp}_{up,down}.csv	Over-representation results for increased and decreased Robust genes: MSigDB Hallmark and GO Biological Process
samples.csv	The 24 analyzed samples: GEO accession, title, group, tissue score, flag
qc_library_metrics.csv	Reads and genes detected per sample
qc_pca.csv	First three principal components per sample
marker_expression.csv	Per-sample expression, $\log_2(\text{CPM} + 1)$, of the tissue-signature and airway marker genes shown in the figures
gene_annotation.csv	Mouse gene symbols and human orthologs used
summary.json	Summary counts and key values used in this report

8.5 Code repository

The complete analysis code, environment specification and instructions for reproducing this report are available at:

https://github.com/Phylomic/phylomic-rnaseq-flagship

Appendix

A. Samples analyzed

The 24 libraries included in this analysis, with their GEO sample records. Reads and genes detected are from the deposited count matrix; the tissue score is defined in Section 7.3, and samples with a score of 6 or above were flagged.

GEO sample	Title	Library	Group	Reads	Genes detected	Tissue score	Flagged
GSM9078227	Lung, CTRL Vehicle, Rep 11	21590	Control	10.0M	22,894	1.5	No
GSM9078228	Lung, CTRL Vehicle, Rep 12	21591	Control	9.0M	22,179	0.6	No
GSM9078229	Lung, CTRL Vehicle, Rep 13	21592	Control	9.6M	23,253	8.8	Yes
GSM9078230	Lung, CTRL Vehicle, Rep 14	21593	Control	9.6M	23,012	9.0	Yes
GSM9078231	Lung, CTRL Vehicle, Rep 15	21594	Control	9.7M	22,842	1.7	No
GSM9078232	Lung, CTRL Vehicle, Rep 16	21595	Control	10.0M	23,322	8.0	Yes
GSM9078233	Lung, CTRL Vehicle, Rep 17	21596	Control	9.9M	22,735	4.8	No
GSM9078234	Lung, CTRL Vehicle, Rep 18	21597	Control	7.5M	22,619	9.2	Yes
GSM9078235	Lung, CTRL Vehicle, Rep 19	21598	Control	10.4M	22,496	1.9	No
GSM9078236	Lung, CTRL Vehicle, Rep 20	21599	Control	10.2M	22,716	4.5	No
GSM9078237	Lung, Single dose BLEO-IPF Vehicle, Rep 1	21600	Bleomycin	9.7M	22,899	1.4	No
GSM9078238	Lung, Single dose BLEO-IPF Vehicle, Rep 2	21601	Bleomycin	10.6M	23,108	0.8	No
GSM9078239	Lung, Single dose BLEO-IPF Vehicle, Rep 3	21602	Bleomycin	10.4M	22,962	0.7	No
GSM9078240	Lung, Single dose BLEO-IPF Vehicle, Rep 4	21603	Bleomycin	11.0M	23,316	1.4	No
GSM9078241	Lung, Single dose BLEO-IPF Vehicle, Rep 5	21604	Bleomycin	11.2M	23,801	1.9	No
GSM9078242	Lung, Single dose BLEO-IPF Vehicle, Rep 6	21605	Bleomycin	8.0M	22,621	9.6	Yes
GSM9078243	Lung, Single dose BLEO-IPF Vehicle, Rep 7	21606	Bleomycin	9.2M	22,389	1.3	No
GSM9078244	Lung, Single dose BLEO-IPF Vehicle, Rep 8	21607	Bleomycin	9.3M	22,413	1.2	No
GSM9078245	Lung, Single dose BLEO-IPF Vehicle, Rep 9	21608	Bleomycin	10.0M	22,903	8.0	Yes
GSM9078246	Lung, Single dose BLEO-IPF Vehicle, Rep 10	21609	Bleomycin	9.7M	23,132	0.9	No
GSM9078247	Lung, Single dose BLEO-IPF Vehicle, Rep 11	21610	Bleomycin	10.5M	23,831	8.9	Yes
GSM9078248	Lung, Single dose BLEO-IPF Vehicle, Rep 12	21611	Bleomycin	9.9M	23,586	1.7	No
GSM9078249	Lung, Single dose BLEO-IPF Vehicle, Rep 13	21612	Bleomycin	9.3M	22,683	1.1	No
GSM9078250	Lung, Single dose BLEO-IPF Vehicle, Rep 14	21613	Bleomycin	7.4M	21,308	0.8	No

B. Supplementary results files

The complete results tables listed in Section 8.4, covering every tested gene and every gene set, accompany this report as supplementary files and are available in the code repository (Section 8.5).